

10-Week Development Plan

Echo: AI-Based Hazardous Sound Detection and Smart Emergency Guidance System

Week 1 — Literature Survey

Objective	Study existing work in sound event detection, TinyML, edge AI, audio scene classification and emergency response systems, and position the project against it.
Result	A written literature review, a clearly defined problem statement, and an identified research gap that the project will address.

Week 2 — Dataset Collection

Objective	Collect and organise public audio datasets (AudioSet, UrbanSound8K, ESC-50, FSD50K) and separate sound classes into dangerous and safe categories.
Result	A consolidated, labelled dataset of dangerous and safe sound classes, ready for preprocessing.

Week 3 — Data Preprocessing

Objective	Convert raw audio into Mel Spectrogram and MFCC representations, and apply augmentation (noise addition, pitch shift, time stretch) to increase data variety.
Result	A cleaned and augmented training dataset in a format ready to be fed into the model.

Week 4 — Train AI Model

Objective	Train a CNN on the prepared dataset and evaluate it using standard classification metrics.
Result	A trained CNN model with recorded accuracy, precision, recall, F1 score and confusion matrix.

Week 5 — Context Verification Module

Objective	Design logic to distinguish real emergencies from false triggers such as a movie, a Bluetooth speaker, running, or location context.
Result	A context-verification module that reduces false positives before an alert is raised.

Week 6 — Emergency Decision Engine

Objective	Design a rule engine that maps a detected sound to a recommended action, safety tips, and relevant emergency numbers.
Result	A working decision engine that converts a classified sound into a concrete, actionable response.

Week 7 — Mobile Application

Objective	Develop a Flutter application with microphone access, push notifications, GPS, and Google Maps integration.
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Result	A functional mobile app front-end with the core UI and supporting features in place.
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Week 8 — AI Integration

Objective	Convert the trained model to TensorFlow Lite and integrate it into the app for on-device, real-time inference.
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Result	Real-time sound detection running locally on the device, with inference speed optimised.
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Week 9 — Testing

Objective	Test the system against gunshots, sirens, movie/TV audio and traffic noise, and measure system performance.
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Result	Recorded results for latency, battery consumption, false positive rate and false negative rate.
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Week 10 — Finalization

Objective	Consolidate the project into a presentable form for submission and demonstration.
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Result	Final report, presentation slides, poster, demo video and complete project documentation.
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